

SCSIT

IV Semester

Database

Exc. 3

- 1. Installing SQL Server**
 - 2. SSMS introduction**
 - 3. Basic DB Administration**
 - 3. Data Insertion**
-

SQL Server and SSMS

☐ SQL Server and SSMS installation

- [Step By Step Installation Of Microsoft SQL Server On Windows System](#)

☐ Connecting to SQL Server

☐ **Standard Security**

(Server=myServerAddress;Database=myDataBase;User
Id=myUsername;Password=myPassword;)

☐ **Trusted Connection**

(Server=myServerAddress;Database=myDataBase;Trusted_Connection=True;)

☐ **Connection to a SQL Server instance**

(Server=myServerName\myInstanceName;Database=myDataBase;User
Id=myUsername;Password=myPassword;)

☐ **Using a non-standard port (default TCP 1433)**

(Server=myServerName,myPortNumber;Database=myDataBase;User
Id=myUsername;Password=myPassword;)

SQL Server and SSMS

Default or Named Instance

SQL Server 2019 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

☐ Default instance
☒ Named instance: MSSQLSERVER

Instance ID: MSSQLSERVER

SQL Server directory: C:\Program Files\Microsoft SQL Server\MSSQL15.MSSQLSERVER

Installed instances:

Instance Name	Instance ID	Features	Edition	Version

< Back Next > Cancel

SQL Server and SSMS

Windows authentication or Mixed mode (SQL Server + Windows authentication)

The screenshot shows the 'SQL Server 2019 Setup' window, specifically the 'Database Engine Configuration' step. The left sidebar lists the installation steps: Install Rules, Feature Selection, Feature Rules, Instance Configuration, Server Configuration, **Database Engine Configuration** (selected), Feature Configuration Rules, Ready to Install, Installation Progress, and Complete. The main area has tabs for 'Server Configuration', 'Data Directories', 'TempDB', 'MaxDOP', 'Memory', and 'FILESTREAM'. The 'Server Configuration' tab is active, showing options for authentication mode. The 'Authentication Mode' section has two radio buttons: 'Windows authentication mode' (unselected) and 'Mixed Mode (SQL Server authentication and Windows authentication)' (selected). Below this, there are two password fields labeled 'Enter password:' and 'Confirm password:', both containing masked characters. A section titled 'Specify SQL Server administrators' contains a list box with one entry: 'cloud-desire-01\Semaphore (Semaphore)'. To the right of the list box, a text box states: 'SQL Server administrators have unrestricted access to the Database Engine.' At the bottom of the list box are three buttons: 'Add Current User', 'Add...', and 'Remove'. The bottom of the window has three navigation buttons: '< Back', 'Next >', and 'Cancel'.

SQL Server 2019 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

Server Configuration | Data Directories | TempDB | MaxDOP | Memory | FILESTREAM

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode

☒ Mixed Mode (SQL Server authentication and Windows authentication)

Specify the password for the SQL Server system administrator (sa) account.

Enter password: [masked]

Confirm password: [masked]

Specify SQL Server administrators

cloud-desire-01\Semaphore (Semaphore)

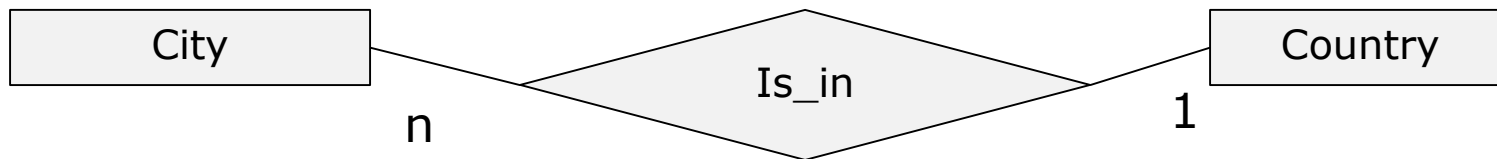
SQL Server administrators have unrestricted access to the Database Engine.

Add Current User | Add... | Remove

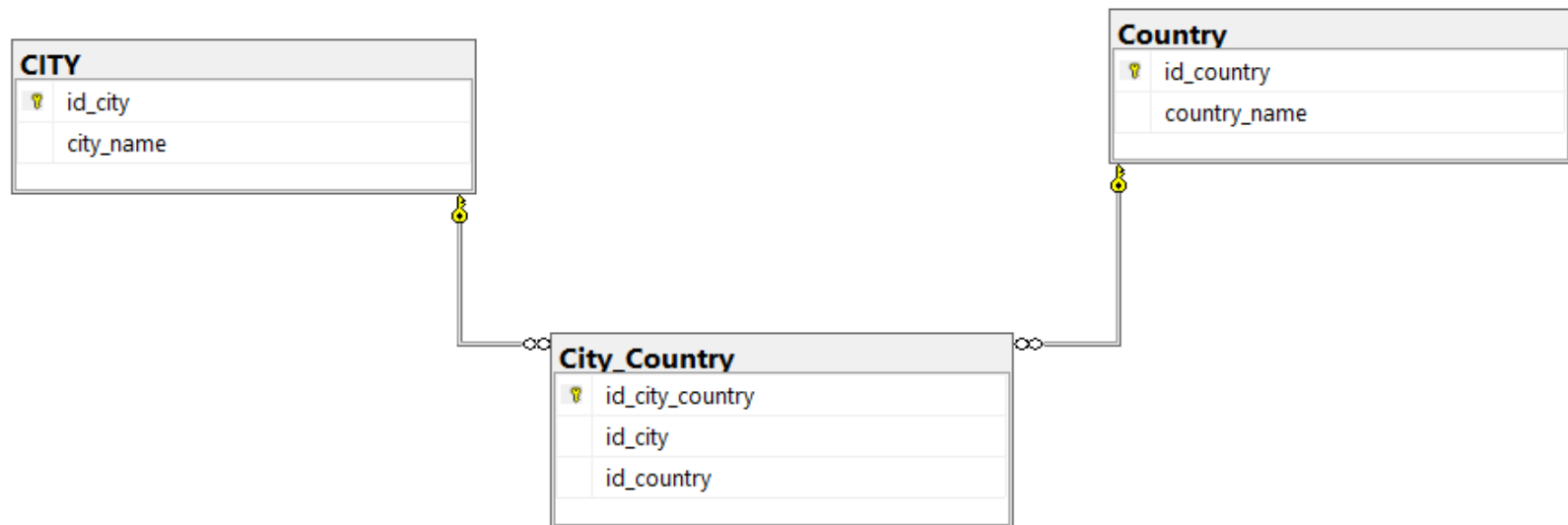
< Back | Next > | Cancel

Ass. 1

- ❑ Create ER diagram (model) with
 - entities
 - ❑ CITY (id_city, city_name)
 - ❑ Country (id_country, country_name)
 - relation
 - ❑ City –***is in*** – Country



- ❑ Create DB based on this model
 - ❑ Analysis of the done activity
-



Database creation Plan

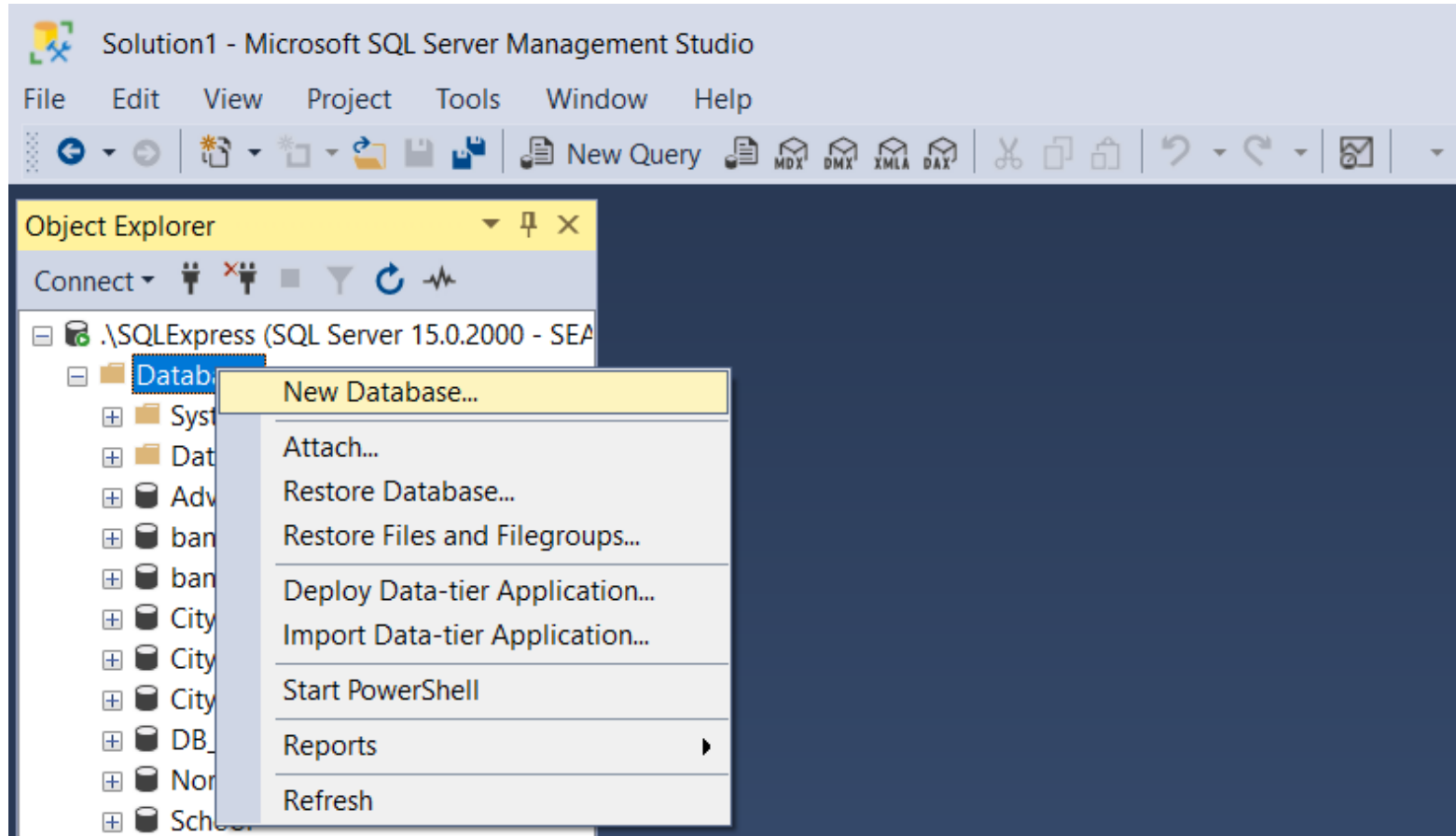
1. Create tables for entities-nomenclatures
2. Create tables for entities that are in relation with some nomenclature
3. Create relationship between 1 and 2 tables.
4. Create tables that are in relation with some of the tables from 1 and 2

....

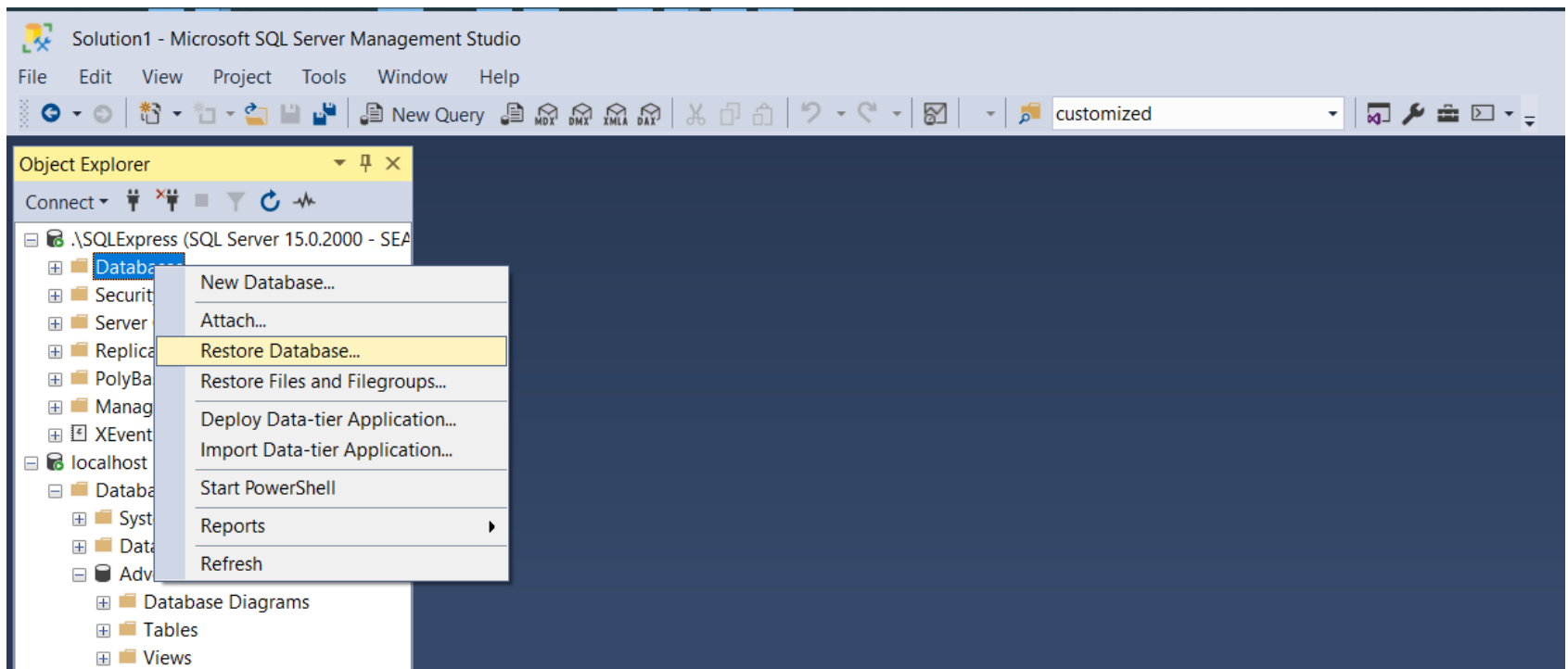
Creation plan

- ❑ **For every entity** (table model), a database table is created with
 - Name (Latin alphabet)
 - Attributes (columns) with appropriate data type
 - ❑ In case with Primary Key, good practice is this attribute to be defined as **auto number/ Identity**. Thanks to this, you allow unique values to be assign for this attribute for every table record
 - **Secondary keys** **MUST** have same or common data type with referenced primary key data type.
 - PK (Text) ➔ FK (Text)
 - PK (Number) ➔ FK (Number)
 - PK (AutoNumber) ➔ FK (Number)
- ❑ Name them associatively

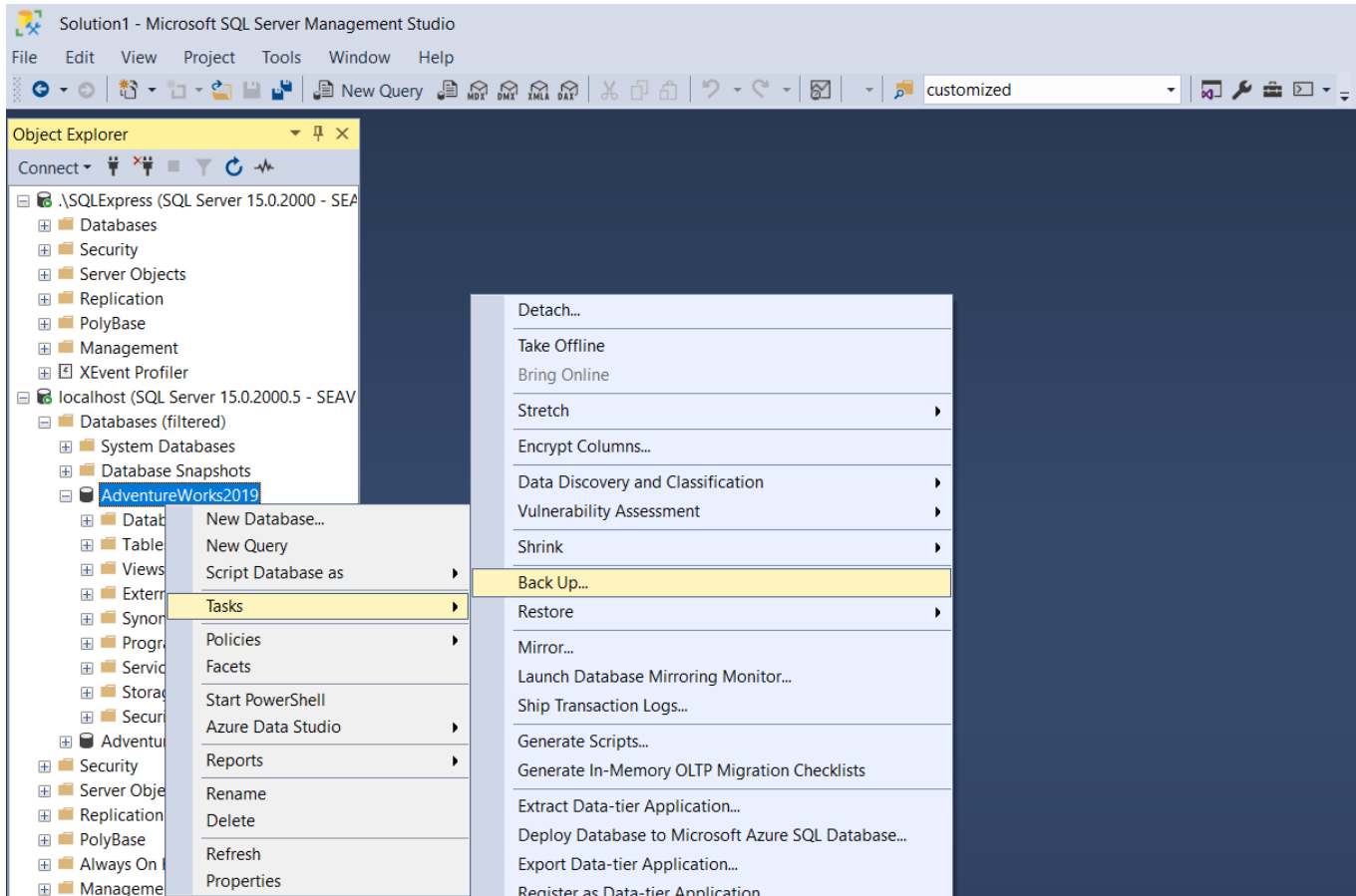
Create database



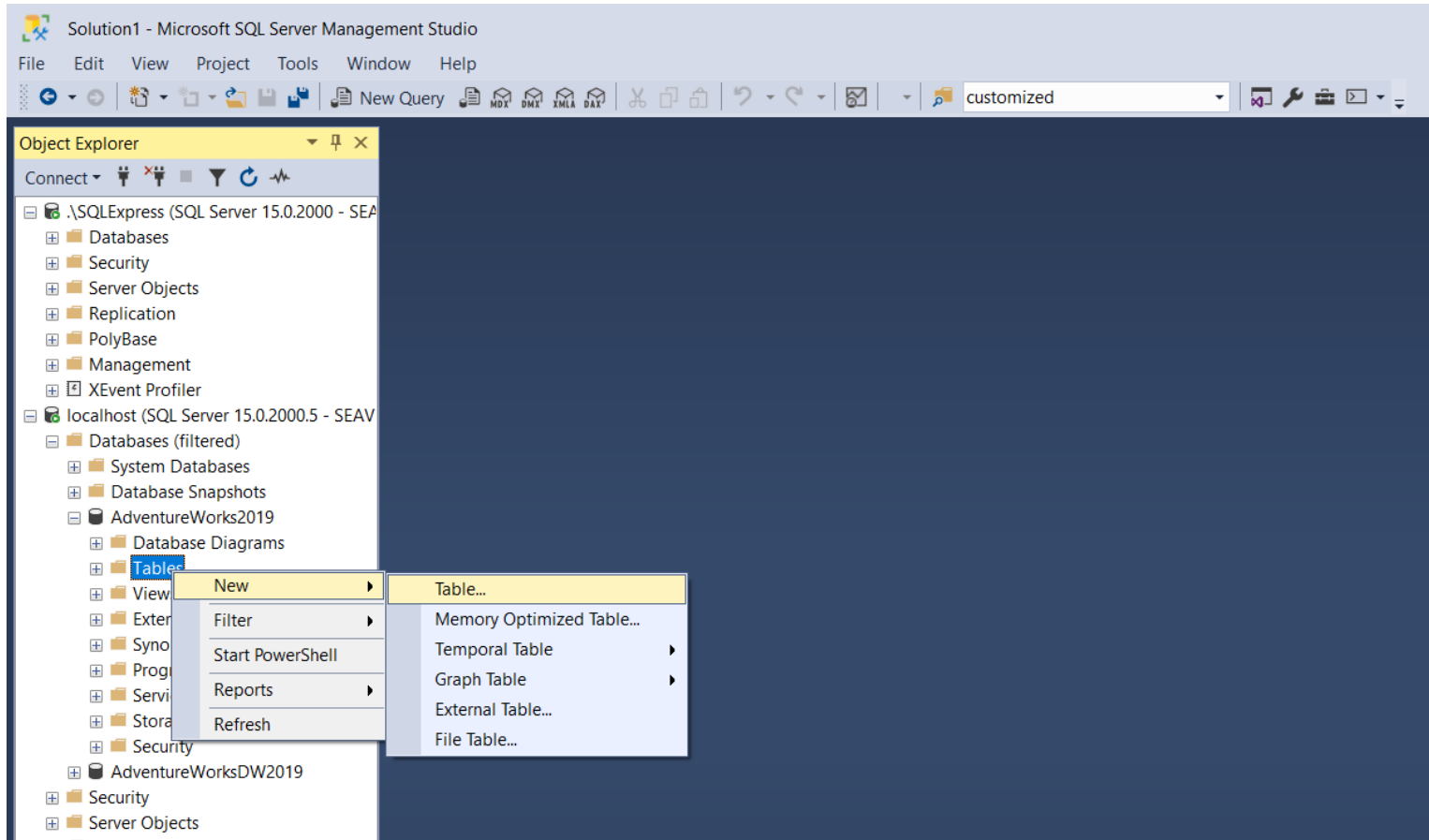
Restore database



Backup database

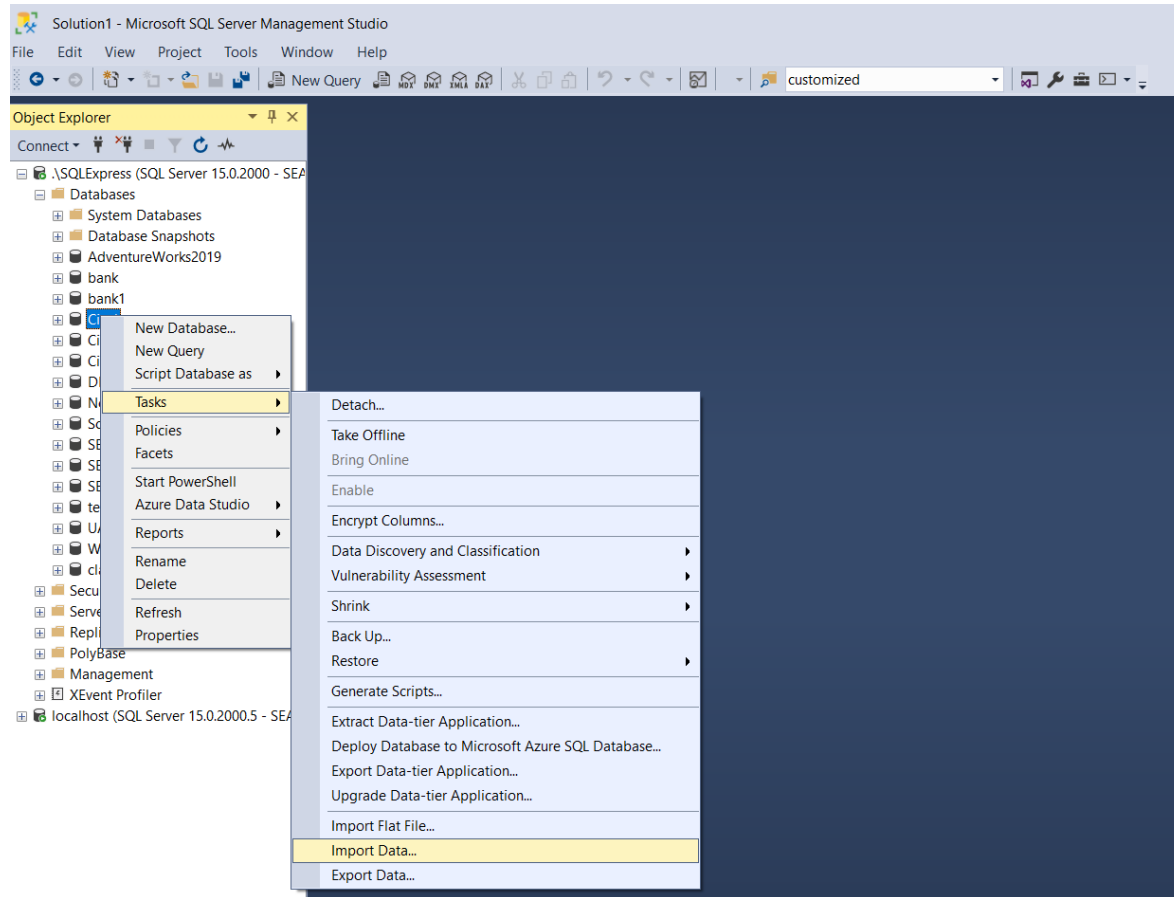


Create table



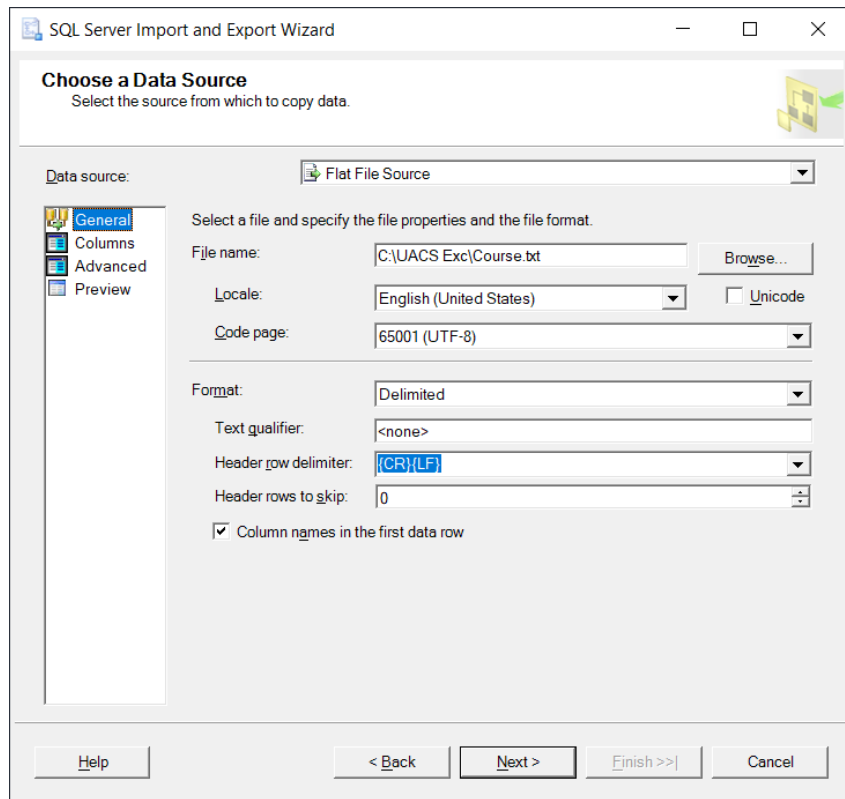
Import data

from source to destination



Import data

from source to destination



SQL Server Import and Export Wizard

Choose a Data Source
Select the source from which to copy data.

Data source: Flat File Source

Select a file and specify the file properties and the file format.

File name: C:\UACS Exc\Course.txt [Browse...](#)

Locale: English (United States) ☐ Unicode

Code page: 65001 (UTF-8)

Format: Delimited

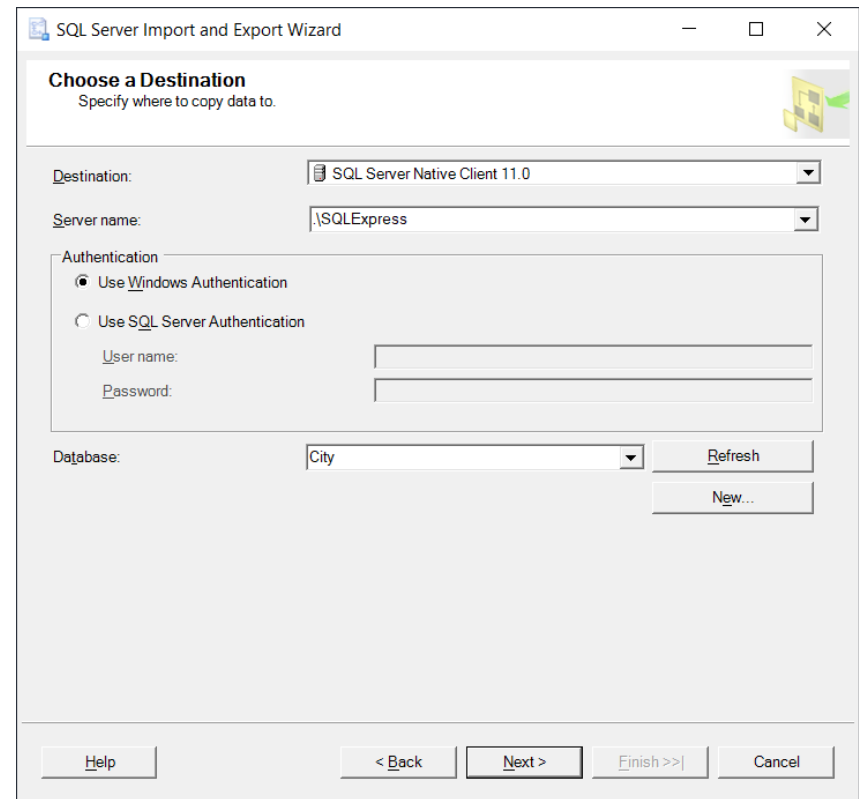
Text qualifier: <none>

Header row delimiter: (CR)(LF)

Header rows to skip: 0

☒ Column names in the first data row

[Help](#) [< Back](#) [Next >](#) [Finish >>](#) [Cancel](#)



SQL Server Import and Export Wizard

Choose a Destination
Specify where to copy data to.

Destination: SQL Server Native Client 11.0

Server name: .\SQLEXPRESS

Authentication

☒ Use Windows Authentication

☐ Use SQL Server Authentication

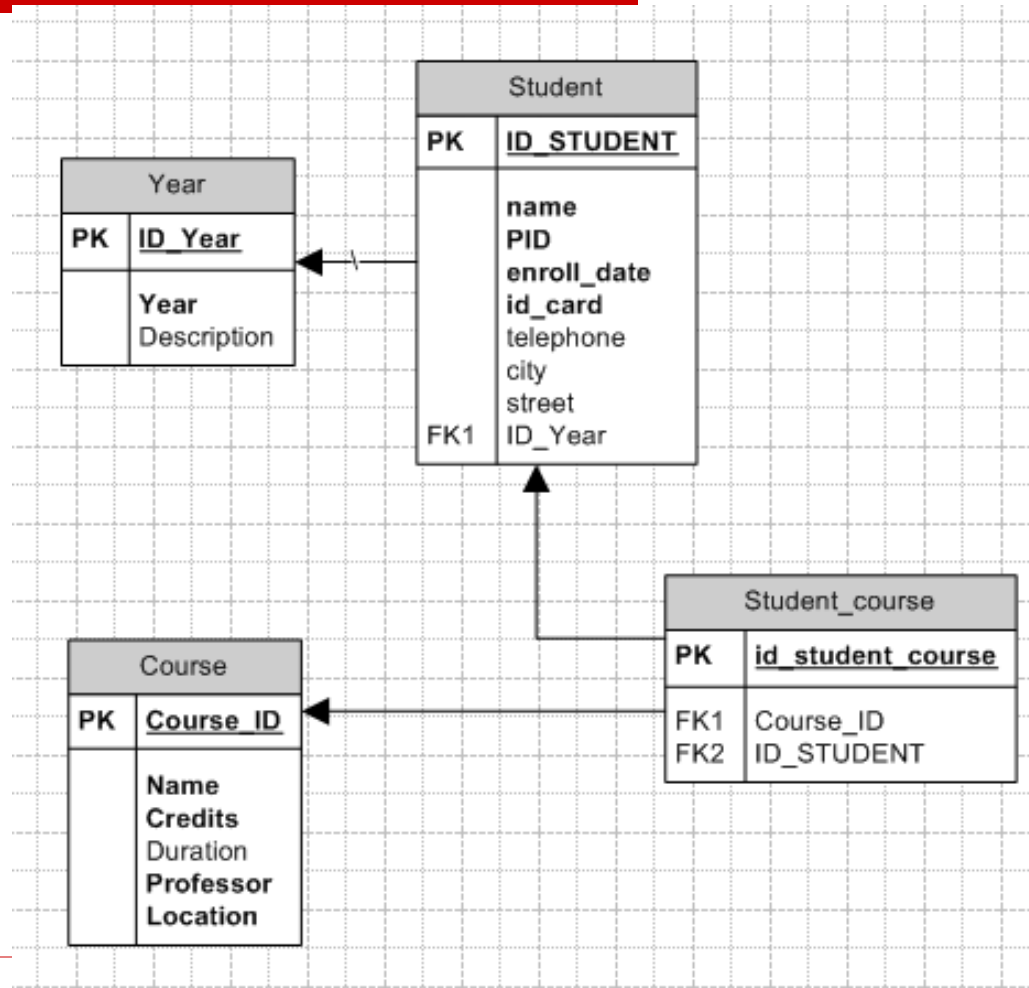
User name:

Password:

Database: City [Refresh](#) [New...](#)

[Help](#) [< Back](#) [Next >](#) [Finish >>](#) [Cancel](#)

Creating DB - Students



Use the model with student- course- year – student_course